NOVA IMS INFORMATION MANAGEMENT SCHOOL

Mini Test I

Computation I / Introduction To Programming

October 24th, 2026

Version I

Read the following instructions carefully:

1. Write your full name, student number and course at the end of this page in a clear and readable manner.
2. Anonymous Tests or tests where the student name is not readable will not be evaluated.
3. The mini test has a duration of 1 hour and 30 minutes. The ending time of the test will be written by the professor on the board before the test begins.
4. The score of each question varies according to the group to which it belongs.
5. If the professor cannot read the student's answer due to poor calligraphy or the choice of more than one answer occurred, the answer will be considered wrong. It is the student's responsibility to type in an understandable manner.
6. **The answers to the multiple choice questions (Group 1) must be marked on the separate Answer Sheet provided to you.** Answers marked only in this document will not be considered: the answers on the Answer Sheet are the final ones.
7. If you want to change an answer on the Answer Sheet, or if you accidentally make any mark on it that could prevent it from being read by an automatic scanner, ask for a new Answer Sheet and fill it in again from scratch. Only clean Answer Sheets, whose answers can be read by an automatic scanner, will be graded.
8. All other answers must be written on the pages of this document. No other paper or document will be evaluated unless it is previously approved and signed by one of the professors.
9. No documents (books, notes, etc...) are allowed to be used during the exam (no computers, smartwatches, phones, etc.).
10. No communication with other students is allowed during the test.
11. All phones must be turned off before the beginning of the test. Students are not allowed to have their phones in their possession even if turned off. Same applies for smart watches and other digital devices.
12. The breaking of any of the previously mentioned rules will cause for you to be expelled from the room, invalidating the mini-test and the necessary disciplinary measures will be applied.

Complete Name: _____

Student number: _____

Practical class: ☐ P1 ☐ P2 ☐ P3 ☐ P4 ☐ P5 ☐ P6 ☐ P7 ☐ ISPA

Exam room: ☐ A120 ☐ Annex 1 ☐ Annex 2 ☐ INE

Group 1

6 points: 1 point for each question.

1. The Rings of Power are numbered from 1 to 20, and rings 4 to 10 were given to the Dwarves. The user inserts the number of a ring. What should be written in place of `<a>` so that the program prints `Dwarves` for rings 4 to 10 (both included) and `someone else` for every other ring?

```
ring = int(input("Which ring? "))
if <a>:
    print("Dwarves")
else:
    print("someone else")
```

A- ◯ `ring >= 4 and ring <= 10`

B- ◯ `ring >= 4 or ring <= 10`

C- ◯ `ring >= 4 and <= 10`

D- ◯ `ring == 4 and ring == 10`

E- ◯ `ring => 4 and ring =< 10`

F- ◯ `ring > 4 and ring < 10`

G- ◯ An error occurs

H- ◯ None of the alternatives

2. Consider the following code. What will be printed on the console if the user inserts `9` (`v1`), then `98` (`v2`) and then `7` (`z`)?

```
v1 = input("How many gold dragons? ")
v2 = input("How many silver stags? ")

z = input("Gimme your fav number: ")

code = int(v1 + v2) + len(z)

if code in [42, 300, 999]:
    print(f"Vault {code} is open. A Lannister always pays his debts.")
elif code > 1000:
    print("Winter is coming.")
else:
    print("You know nothing, Jon Snow.")
```

A- ◯ `Vault 1005 is open. A Lannister always pays his debts.`

B- ◯ `Vault 108 is open. A Lannister always pays his debts.`

C- ◯ `You know nothing, Jon Snow.`

D- ◯ `Vault {code} is open. A Lannister always pays his debts.`

E- ◯ `Winter is coming.`

F- ◯ `Vault 999 is open. A Lannister always pays his debts.`

G- ◯ An error occurs

H- ◯ None of the alternatives

3. The user inserts an integer number. What should be written in place of `<a>` so that the program prints `even` when the number is even and `odd` when it is odd?

```
n = int(input("Gimme an integer number: "))
if <a>:
    print("even")
else:
    print("odd")
```

A- ◯ `n / 2 == 0`

B- ◯ `n % 2 == 0`

C- ◯ `n // 2 == 0`

D- ◯ `n % 2 == 1`

E- ◯ `n ** 2 == 0`

F- ◯ `n % 2`

G- ◯ An error occurs

H- ◯ None of the alternatives

4. Consider the following code. What will be printed on the console?

```
points = 70
if points > 50:
    if points > 150:
        print("Gryffindor caught the Snitch!")
else:
    print("Slytherin wins the match!")
```

A- ◯ `points`

B- ◯ `Gryffindor caught the Snitch!`

C- ◯ Nothing is printed

D- ◯ `70`

E- ◯ `Slytherin wins the match!`

F- ◯ `Gryffindor caught the Snitch!`
`Slytherin wins the match!`

G- ◯ An error occurs

H- ◯ None of the alternatives

5. What should be written in place of `<a>`, `<b>` and `<c>` so that `1 4 7 10 13 16 19` is printed on the console?

```
for i in range(<a>, <b>, <c>):
    print(i, end=" ")
```

A- ◯ `<a>` = 1, `<b>` = 19 and `<c>` = 3

B- ◯ `<a>` = 1, `<b>` = 20 and `<c>` = 4

C- ◯ `<a>` = 1, `<b>` = 20 and `<c>` = 3

D- ◯ `<a>` = 3, `<b>` = 20 and `<c>` = 1

E- ◯ `<a>` = 1, `<b>` = 3 and `<c>` = 20

F- ◯ `<a>` = 0, `<b>` = 20 and `<c>` = 3

G- ◯ An error occurs

H- ◯ None of the alternatives

6. Consider the following code. What will be printed on the console?

```
finds = "rex-raptor-REX-bronto-rex"
lt = finds.split("-")
res = 0
for x in lt:
    if x == "rex":
        res = res + 1
print(f"{res} of {len(lt)} fossils are rex")
```

A- ◯ `3 of 5 fossils are rex`

B- ◯ `2 of 25 fossils are rex`

C- ◯ `5 of 5 fossils are rex`

D- ◯ `{res} of {len(lt)} fossils are rex`

E- ◯ `0 of 1 fossils are rex`

F- ◯ `2 of 5 fossils are rex`

G- ◯ An error occurs

H- ◯ None of the alternatives

Group 2

4 points: 2 points for each question.

NOTE: Each question in this group shows a block of code and states what that code was supposed to do. There is an issue with each block of code: it does not do what was intended. Your task is to find the issue and fix it, as follows:

A State the issue briefly on the line after "What is the issue?".

B Rewrite each faulty line of code, corrected, on the empty line directly below it. Leave all the other empty lines blank.

Example of a complete answer. The answers are written in blue, and the letters in the margin show where each step was answered.

This block of code was supposed to print `yayyy yummy!` when the favourite fruit of the user is grapefruit, however it is capitalised.

A **What is the issue?** *print() returns nothing; () missing after lower*

Correctly identify and **fix the problematic line(s)** below.

```
1 fruit = print(input("Gimme your favourite fruit?"))
```

B *fruit = input("Gimme your favourite fruit?")*

```
2
3 if fruit.lower == "grapefruit":
```

B *if fruit.lower() == "grapefruit":*

```
4     print("yayyy yummy!")
```

The questions of this group start on the next page.

1. This block of code was supposed to warn when the first number of the list is smaller than 5 and, otherwise, to say whether the list has more than three numbers.

What is the issue? _____

_____

_____

Correctly identify and **fix the problematic line(s)** below.

```
1 lt = [10, 20, 30]
2
3 if len(lt[0]) < 5:
4     print(f"warning! first number too small")
5 elif len(lt) > 3:
6     print(f"{lt} has more than 3 elements, all good!")
7 else:
8     print(f"{lt} has 3 elements or fewer, ew...")
```

2. This block of code was supposed to create the Pawnee Parks and Recreation email of each member of the staff and store them in a list. When the code runs, exactly the following was supposed to be printed on the console, and nothing else:
`['leslie@pawnee.gov', 'ron@pawnee.gov', 'april@pawnee.gov']`

What is the issue? _____

_____

_____

Correctly identify and **fix the problematic line(s)** below.

```
1 staff = ["Leslie", "Ron", "April"]
2 res = []
3 for x in staff:
4     res.append(x.lower() + "@pawnee.gov")
5     print(res)
```

Group 3

2 points

NOTE: This group corresponds to direct questions. Please answer the questions within the space provided for the answer. The amount of space provided is just speculative. Do not feel inclined to add more just to fill in the space.

1. The string `family` below contains the names of the Griffin family, separated by semicolons. What needs to be written instead of `<A>`, `<B>`, `<C>`, `<D>` and `<E>` so that the program works as described below?

   - The string is split into a list with one name in each position (`<A>`).
   - The loop goes through every name of that list (`<B>` and `<C>`).
   - Each name with more than four letters (`<D>`) is added to `res` in capital letters (`<E>`).

   In the end, the program prints `['PETER', 'CHRIS', 'STEWIE']`.

```
family = "peter;lois;meg;chris;stewie"
lt = family.split(<A>)
res = []

for <B> in <C>:
    if <D>:
        res.append(<E>)

print(res)
```

(0.4 points `<A>`): _____

(0.4 points `<B>`): _____

(0.4 points `<C>`): _____

(0.4 points `<D>`): _____

(0.4 points `<E>`): _____

Group 4

8 points

NOTE: This group corresponds to open questions. Make sure you present all necessary code when code is requested from you. Feel free to write notes and other things to show your thought process when you feel like it may be necessary.

Teddy, the most loyal customer of Bob's Burgers, has just sat down at the counter. The program starts with the following two lines of code, which are already written:

```
menu = ["bacon", "cheese", "teriyaki", "mushroom", "pineapple", "beets"]
orders = ["fries", "coleslaw"]
```

Continue the program, answering each of the following questions below the previous one.

1. **(5 points)** Write the code that does the following:

   a) Print `Today's menu is:` followed by the menu, one burger per line.

   b) Ask Teddy which burger he would like.

   c) If that burger is on the menu (regardless of the capitalisation Teddy used to order it), add it to Teddy's orders and print a message with Teddy's full order saying that it is coming right up. Otherwise, print a message saying that it is not on the menu.

   Example of a correct run of the program (what Teddy inserts is in grey; your messages may be worded differently):

   ```
   Today's menu is:
   bacon
   cheese
   teriyaki
   mushroom
   pineapple
   beets
   What burger would you like, Teddy? BEETS
   Teddy's order: ['fries', 'coleslaw', 'beets']. Coming right up!
   ```

   Write your answer in the box on the next page.

2. **(3 points)** Teddy's fries and coleslaw are part of a combo that costs 5 euros. All burgers cost 8.99 euros, but every day, the last burger on the menu has a 10% discount. Calculate the total cost of Teddy's order, add 23% tax to it and print the final total, rounded to two decimal places. In this question, you can assume that the burger Teddy ordered is on the menu. **Your code must work for any day, even if the menu changes.**

   Example of a correct run of the program, continuing the example of the previous question (Teddy ordered the beets burger, which happens to be the last one on today's menu. Your message may be worded differently as long as it conveys the intended message):

   ```
   With tax, Teddy's total is 16.1 euros.
   ```

   Write your answer in the box below.

**** EXTRA PAGE ****

Good work! ☺